AIGP Module 7: Other Laws and Standards Related to AI

Comprehensive Study Notes

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1. INTRODUCTION TO MODULE 7

1.1 Purpose and Scope

Purpose: AI governance professionals must understand the broader legal landscape beyond AI-specific regulations. AI systems are subject to numerous existing laws that weren't designed for AI but nevertheless apply.

Key Principle: Legal compliance is a critical component of AI governance and risk management.

Module Coverage:

1. **AI-Specific Laws and Standards**: Regulations and frameworks specifically designed for AI
2. **Data Privacy Laws**: How privacy regulations apply to AI
3. **Other Existing Laws**: IP, non-discrimination, consumer protection, product liability

Why This Matters:

- AI doesn't operate in legal vacuum
- Multiple legal regimes apply simultaneously
- Non-compliance carries serious consequences
- Understanding legal landscape essential for governance

1.2 Multi-Jurisdictional Challenge

Global Variation:

- No universal AI regulation
- Different approaches across jurisdictions
- Organizations must navigate multiple legal systems
- Alignment and dissonance between frameworks

Compliance Strategy:

- Identify all applicable jurisdictions
- Understand requirements in each
- **Harmonize to strictest requirements** (most common approach)

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- Develop unified compliance framework
- Remain alert and prepared to adjust

Brussels Effect: EU regulations (like GDPR, EU AI Act) often become de facto global standards

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2. AI-SPECIFIC LAWS AND STANDARDS

2.1 Global AI Regulation Landscape

2.1.1 European Union

EU AI Act (covered extensively in Module 6):

- First comprehensive AI regulation
- Risk-based approach
- Extraterritorial reach
- Entered force August 2024, phased implementation

Key Features:

- Four risk categories (Prohibited, High-Risk, Limited Risk, Minimal)
- Strict requirements for high-risk AI
- GPAI regulation
- Significant penalties

Impact: Setting global precedent; many organizations worldwide must comply

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2.1.2 United States

Federal Level:

No Comprehensive Federal AI Law: U.S. lacks single overarching AI regulation (as of now)

Executive Orders and Federal Guidance:

Executive Order 14110 (October 2023): "Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence"

- Directs federal agencies on AI governance
- Safety testing for powerful AI models
- Equity and civil rights protections
- Standards development
- Innovation support

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Executive Order 13960 (December 2020): "Promoting the Use of Trustworthy Artificial Intelligence in the Federal Government"

- Principles for federal AI use
- Reliability, transparency, explainability

AI Bill of Rights Blueprint (October 2022):

- Non-binding guidance
- Five principles:
    1. Safe and effective systems
    2. Algorithmic discrimination protections
    3. Data privacy
    4. Notice and explanation
    5. Human alternatives, consideration, and fallback

OMB Guidance (March 2024): Memorandum to federal agencies

- Advance AI governance, innovation, and risk management
- Requirements for federal agencies' AI use
- Safety, security, rights protections
- Transparency and accountability

Department of Homeland Security (DHS) Roles and Responsibilities Framework for AI in Critical Infrastructure:

- Guidance for critical infrastructure sectors
- Risk management for AI

Sectoral Regulation:

Federal Trade Commission (FTC):

- Enforces against unfair and deceptive practices
- Applies existing authority to AI
- Focus on algorithmic bias, privacy violations, false claims
- Statements that existing consumer protection laws apply to AI

Consumer Financial Protection Bureau (CFPB):

- Oversees AI in financial services
- Fair lending laws (ECOA, FCRA)
- Adverse action notice requirements
- Explainability expectations

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Equal Employment Opportunity Commission (EEOC):

- Enforces employment discrimination laws
- Guidance on AI in hiring and employment
- Title VII applicability to algorithmic discrimination

Food and Drug Administration (FDA):

- Regulates AI/ML-based medical devices
- Risk-based framework for software as medical device (SaMD)
- Emphasis on continuous learning and updates

Federal Reserve / OCC (Banking Regulators):

- Model Risk Management (SR 11-7)
- Governance for models in banking
- Validation, testing, ongoing monitoring

State and Local AI Laws:

Why State/Local Action?

- Absence of federal legislation
- States acting to fill gap
- Variation across jurisdictions

Common Themes:

- Transparency and disclosure
- Right to opt out of automated decision-making
- Bias audits
- Prohibitions on certain uses

Key State Laws:

California:

AB 2013: Generative AI Training Data Transparency

- Requires disclosure of training data sources for generative AI
- Transparency for copyrighted content use
- Effective date provisions for implementation

SB 942: AI Transparency Act

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- Requirements for disclosure of AI use
- Particularly for generative AI

Other California Laws:

- CCPA/CPRA (privacy laws with implications for AI)
- Automated decision-making provisions

Colorado:

SB 24-205: Colorado AI Act

- Consumer protections from algorithmic discrimination
- Risk management requirements
- Impact assessments
- Transparency and disclosure
- Right to opt out
- Developer and deployer obligations
- Enforcement by Colorado Attorney General

Key Features:

- Broad scope (many AI systems covered)
- Focus on high-risk decisions
- Algorithmic impact assessments required

Utah:

SB 149: AI Policy Act

- Regulates government use of AI
- Requirements for state agencies
- Transparency, accountability, human oversight

New York City:

Local Law 144: Bias Audits of Automated Employment Decision Tools (AEDT)

- **Effective**: July 2023
- **Requirement**: Employers using AI for hiring/promotion must conduct annual bias audits
- **Bias Audit**: Independent analysis by third party or internal auditor

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- **Metrics**: Test for disparities across race/ethnicity and gender
- **Publication**: Audit results must be published on website
- **Notice**: Candidates must be notified of AI use and their rights
- **Penalties**: Fines for non-compliance

Significance: First law mandating bias audits for employment AI; model for other jurisdictions

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Other States:

- Illinois: Biometric Information Privacy Act (BIPA)—implications for facial recognition
- Washington: Proposed legislation on facial recognition
- Multiple states considering AI bills

2.1.3 China

Approach: Government-led regulation with emphasis on security and social stability

Key Regulations:

Cyberspace Administration of China (CAC) Guidelines (2023):

- Generative AI regulations
- Content management and censorship
- Registration and approval requirements
- Algorithmic recommendations regulation

Key Features:

- Rights-based approach (consumer rights, transparency)
- Content control (prevent harmful content)
- Security requirements
- Data protection

Municipal Laws:

- Local governments have additional oversight
- Compliance audits
- Possible bans for safety, discrimination, national security concerns

Social Credit Systems: Government-run but distinct from prohibited social scoring under EU AI Act

Characteristics:

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- National and local systems
- Track behavior and trustworthiness
- Consequences for low scores (travel restrictions, loan denials)
- Integration with AI and big data

2.1.4 Canada

Proposed Federal Legislation:

Bill C-27: Digital Charter Implementation Act (2022)

- Omnibus legislation addressing digital issues

Artificial Intelligence and Data Act (AIDA) (Part of Bill C-27):

- Proposed AI-specific regulation
- Risk-based approach
- High-impact systems subject to requirements
- Assessments, mitigation, transparency
- Prohibitions on certain harmful uses
- Minister of Innovation empowered to oversee

Consumer Privacy Protection Act (CPPA) (also in Bill C-27):

- Updates to privacy law (replaces PIPEDA)
- AI-relevant provisions

Creates New Tribunal: Data protection tribunal for privacy disputes

Voluntary Code of Conduct (September 2023):

- Voluntary commitments by AI companies
- Responsible development of advanced generative AI
- Principles: accountability, safety, fairness, transparency, human oversight

Status: Bill C-27 still under legislative process (as of module creation)

2.1.5 Singapore

Approach: Principles-based, voluntary frameworks with practical tools

Model AI Governance Framework:

- First released 2019, updated for generative AI
- Voluntary guidance

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- Principles: transparency, explainability, fairness, human oversight, accountability
- Risk-based approach
- Implementation guides for organizations

AI Verify:

- Open-source toolkit
- Enables organizations to test AI systems against standards
- Objective assessment of AI performance (bias, robustness, etc.)
- Validation and transparency
- Global adoption encouraged

Characteristics:

- Non-binding (guidance, not law)
- Encourages adoption of best practices
- Supports innovation while promoting responsible AI
- International collaboration

2.1.6 Brazil

Proposed Legislation:

Brazilian AI Bill (under consideration):

- Comprehensive, risk-based framework
- Inspired by EU AI Act
- Three risk levels (similar to EU approach)
- Human oversight requirements
- Rights protections
- Emphasis on transparency and accountability

Status: Legislative process ongoing

Existing Laws:

- LGPD (Lei Geral de Proteção de Dados—Brazilian GDPR equivalent)
- Consumer protection laws
- Civil rights framework

2.1.7 India

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Approach: Developing frameworks with emphasis on digital economy growth

AI Governance Principles:

- Government issued principles for responsible AI
- Safety, security, privacy, transparency, accountability, fairness
- Non-binding guidance

Proposed Digital India Act:

- Comprehensive digital regulation (replacing IT Act 2000)
- Expected to include AI provisions
- Focus on online safety, data protection, algorithmic accountability

Advisories on AI:

- Government advisories on consent for AI training data
- Labeling requirements for AI-generated content
- Deepfake concerns and regulations

Sectoral Regulation:

- Financial services: RBI guidance on AI/ML use
- Healthcare: Emerging AI regulations

2.1.8 Japan

Approach: Non-binding guidelines and national strategy

AI Strategy:

- National AI strategy for economic growth
- Focus on Society 5.0 (super-smart society)
- Investment in AI research and development

Guidelines:

- Non-binding AI guidelines
- Principles: human-centric, appropriate operation, adequate security
- Voluntary adoption

Characteristics:

- Emphasis on innovation and competitiveness
- Self-regulation and industry leadership
- International cooperation

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2.2 Key AI Governance Standards

2.2.1 ISO Standards for AI

ISO/IEC 22989:2022 – AI Concepts and Terminology:

Purpose: Establish common terminology for AI field

Content:

- Defines over 100 key AI concepts
- Standardized vocabulary for AI systems, processes, stakeholders
- Core concepts: AI system, machine learning, training data, model, etc.

Why Important:

- **Harmonization**: Common language across organizations, jurisdictions
- **Clarity**: Reduces confusion and miscommunication
- **Foundation**: Basis for other standards and regulations

Scope: All types of AI systems

Use: Reference document for AI professionals, policymakers, standards developers

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ISO/IEC 42001:2023 – AI Management System:

Purpose: Guidance for establishing, implementing, maintaining, and improving AI management systems

Approach: Integrated management system for AI governance

Content:

- Requirements for AI management system
- Risk-based approach
- Continuous improvement cycle (Plan-Do-Check-Act)
- Policies, processes, documentation
- Roles and responsibilities
- Risk assessment and treatment
- Monitoring and review

Applicability: Organizations of any size, any industry

Certification: Organizations can be certified to ISO 42001 (demonstrates commitment)

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Benefits:

- Structured approach to AI governance
- Internationally recognized
- Integration with other management systems (ISO 27001 for information security, ISO 9001 for quality)
- Demonstrates due diligence

Relationship to EU AI Act: Can help demonstrate compliance with QMS requirements (Article 17)

Other Relevant ISO Standards:

- **ISO/IEC 23894**: Risk management for AI
- **ISO/IEC 27001**: Information security (applicable to AI systems)
- **ISO/IEC 27701**: Privacy information management
- **ISO 31000**: Risk management (general, applicable to AI)

2.2.2 IEEE Standards

IEEE (Institute of Electrical and Electronics Engineers): Professional association developing technical standards

Ethically Aligned Design:

- Framework for ethical AI development
- Principles: human rights, well-being, data agency, effectiveness, transparency, accountability, awareness of misuse

IEEE 7000 Series (Standards on AI Ethics):

- **IEEE 7000**: Model process for addressing ethical concerns during system design
- **IEEE 7001**: Transparency of autonomous systems
- **IEEE 7002**: Data privacy processes
- **IEEE 7003**: Algorithmic bias considerations
- **IEEE 7010**: Well-being impact assessment

Purpose: Provide technical guidance for implementing ethical AI

Use: Voluntary standards for organizations and developers

2.2.3 Council of Europe HUDERIA

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Full Name: Human Rights, Democracy, and the Rule of Law Impact Assessment for AI Systems

Purpose: Framework for assessing AI impacts on fundamental European values

Principles:

- Human dignity
- Freedom and autonomy
- Democracy and rule of law
- Equality and non-discrimination
- Solidarity
- Privacy and data protection

Process:

- Assess potential impacts on rights and values
- Identify and mitigate risks
- Proportionality and necessity assessment
- Stakeholder engagement

Content:

- Risk identification methodology
- Evaluation criteria based on European values
- Mitigation strategies
- Documentation and communication

Use: Particularly for public sector and fundamental rights-sensitive applications

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2.3 AI Governance Framework Development

Organizational Considerations:

When developing AI governance frameworks, organizations must determine:

1. Principles: What values guide AI use? (fairness, transparency, privacy, safety, etc.)

2. Risk Tolerance: How much risk is acceptable? Risk appetite

3. Jurisdiction: Which laws and regulations apply? (based on location, customers, operations)

4. Industry/Sector: Sector-specific requirements and standards

5. AI Relationship to Business Strategy: Is AI core to business? Supporting role? How critical?

6. AI Purpose: What is AI used for? (internal operations, customer-facing, product)

7. Size and Implementation Ability: Resources available (budget, expertise, time)

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Framework Components:

- Policies and procedures for AI lifecycle (from Module 4)
- Governance structures and roles (from Module 3)
- Risk management processes (NIST AI RMF, ISO standards)
- Compliance mechanisms (legal, regulatory)
- Stakeholder engagement
- Monitoring and continuous improvement

AI Processes to Consider:

- **Development**: Building AI in-house
- **Procurement**: Acquiring third-party AI
- **Use/Deployment**: Operating AI systems

Each process needs governance tailored to organizational role

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3. DATA PRIVACY LAWS AND AI

3.1 Why Privacy Laws Matter for AI

AI and Personal Data:

- AI often processes personal data (training, operation)
- Data privacy laws regulate personal data processing
- AI raises specific privacy challenges (scale, inference, profiling)

Key Privacy Laws:

- **GDPR** (EU General Data Protection Regulation)
- **CCPA/CPRA** (California Consumer Privacy Act / Privacy Rights Act)
- **LGPD** (Brazil)
- Many others globally

Intersection: Privacy laws weren't designed for AI but apply fully to AI systems processing personal data

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3.2 Core Privacy Principles Applicable to AI

Fair Information Practices (FIPs) (from Module 2, applied to AI context):

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1. Notice / Awareness:

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Principle: Inform individuals about data collection and use

AI Application:

- Notify individuals that AI uses their data
- Explain how AI processes data
- Describe purposes (training, operation, improvement)
- Disclose data sources

Challenges:

- Complex AI systems hard to explain
- Multiple data uses in AI lifecycle

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2. Choice / Consent:

Principle: Individuals should control their data

AI Application:

- Obtain consent for data use in AI (when required by law)
- Provide meaningful choices (opt-in/opt-out)
- Specific consent for sensitive data or high-risk AI

Challenges:

- Meaningful consent difficult (complexity of AI)
- Consent fatigue
- GDPR: automated decision-making requires explicit consent (with exceptions)

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3. Access / Participation:

Principle: Individuals can view and correct their data

AI Application:

- Right to access personal data used by AI
- Right to correct inaccuracies
- Right to understand AI decisions

Challenges:

- Training data access (feasibility, others' privacy)
- Understanding complex AI outputs

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4. Integrity / Security:

Principle: Data must be accurate and secure

AI Application:

- Data quality for accurate AI (garbage in, garbage out)
- Security against breaches, adversarial attacks
- Protection of training data

AI-Specific Risks:

- Data poisoning attacks
- Model inversion (reconstructing training data)
- Large-scale data processing increases breach impact

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5. Enforcement / Redress:

Principle: Mechanisms to enforce principles and remedy violations

AI Application:

- Right to challenge AI decisions
- Complaints to data protection authorities
- Judicial remedies
- Regulatory enforcement (penalties)

3.3 GDPR and AI

General Data Protection Regulation (EU): Comprehensive privacy law with significant AI implications

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3.3.1 Applicability to AI

When GDPR Applies:

- Processing personal data of individuals in EU
- Offering goods/services to EU residents
- Monitoring behavior of EU residents

Personal Data: Information relating to identified or identifiable natural person

AI Uses of Personal Data:

- Training data containing personal information

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- Inputs during operation (user data)
- Outputs revealing information about individuals
- Profiling and automated decision-making

3.3.2 Automated Decision-Making (Article 22)

Article 22(1): Data subject has right not to be subject to decision based **solely on automated processing**, including profiling, which produces **legal effects** or **similarly significantly affects** them

Key Terms:

- **Solely automated**: No meaningful human involvement
- **Legal effects or similarly significant**: Credit decisions, employment, insurance, benefits eligibility, etc.

Prohibition: Generally prohibited unless exception applies

Exceptions (Article 22(2)): Automated decisions allowed if:

1. **Necessary for contract** performance (entering or performing contract)
2. **Authorized by EU/Member State law** with safeguards
3. **Based on explicit consent** of data subject

Safeguards Required (Article 22(3)):

- Right to obtain human intervention
- Right to express point of view
- Right to contest the decision
- Right to obtain an explanation

Implications for AI:

- High-stakes AI decisions require human involvement OR explicit consent with safeguards
- Must enable human review, challenge, explanation
- Organizations must assess if AI falls under Article 22

Recital 71: Emphasizes need for suitable safeguards, transparency about logic involved, significance and consequences for data subject

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3.3.3 Data Protection Impact Assessment (DPIA) for AI

Article 35: DPIA required when processing likely to result in **high risk** to rights and freedoms

When Required for AI:

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- Systematic and extensive profiling with significant effects (e.g., credit scoring, employment decisions)
- Large-scale processing of special category data (e.g., health AI)
- Systematic monitoring of publicly accessible areas (e.g., facial recognition)

DPIA Process (for AI):

1. Describe processing (AI system, data, purposes)
2. Assess necessity and proportionality
3. Assess risks to data subject rights and freedoms
4. Identify measures to address risks (technical, organizational, PETs)
5. Consult Data Protection Officer
6. Consult data subjects or representatives (where appropriate)
7. Document assessment
8. If high residual risk, consult supervisory authority before processing

Outcome: Decision to proceed, modify, or not proceed with AI system

Governance Integration: DPIA often part of broader impact assessment (AIA from Module 4)

3.3.4 Special Categories of Data (Sensitive Data)

Article 9: Prohibits processing special categories of personal data (with exceptions)

Special Categories (Sensitive Data):

- Racial or ethnic origin
- Political opinions
- Religious or philosophical beliefs
- Trade union membership
- Genetic data
- Biometric data (for unique identification)
- Health data
- Sex life or sexual orientation

General Prohibition: Cannot process special category data

Exceptions (Article 9(2)) - Processing allowed if:

- **Explicit consent** of data subject
- Necessary for **employment, social security, social protection** law

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- Necessary to protect **vital interests** (life-or-death)
- Processing by not-for-profit with **political, philosophical, religious, trade union aims** (limited to members/former members)
- Data **manifestly made public** by data subject
- Necessary for **legal claims or judicial acts**
- Necessary for reasons of **substantial public interest** (with basis in law, proportionate, safeguards)
- Necessary for **health/social care** (by health professional or under confidentiality)
- Necessary for **public health** (official authority or health professional)
- Necessary for **archiving, research, statistics** (with safeguards)

AI Implications:

Bias Testing Dilemma:

- Fairness testing requires demographic data (special categories)
- Data minimization principle discourages collection
- Conflict between bias mitigation and data minimization

Solutions:

- Collect special category data with explicit consent or legal basis (e.g., substantial public interest for bias testing)
- Generate proxies (infer from non-sensitive data—caution: accuracy and ethics)
- Purchase aggregated data (from vendors)
- Ask users voluntarily (for testing, separate from operational data)

Health AI:

- Frequently processes health data
- Must rely on appropriate legal basis (consent, healthcare provision, public health, research)
- Extra safeguards required

3.3.5 Pseudonymization and Anonymization

Definitions:

Anonymization:

- **Irreversible** removal of identifiers so individual cannot be re-identified by **anyone**
- **High threshold** under GDPR (extremely difficult to achieve)

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- If truly anonymized, data is **no longer personal data** → GDPR doesn't apply

Pseudonymization (Article 4(5)):

- **Reversible** replacement of identifiers with pseudonyms
- Re-identification possible with additional information (held separately under security)
- **Still personal data** under GDPR → GDPR applies
- GDPR recognizes as security measure (reduces risk)

AI Context:

Anonymization Benefits:

- AI can train on anonymized data without GDPR restrictions
- Falls outside GDPR scope if truly anonymized

Anonymization Challenges:

- Very hard to achieve (advances in re-identification techniques)
- Large datasets, rich data → easier to re-identify
- GDPR requires guarantees anonymization is irreversible

Pseudonymization for AI:

- Reduces privacy risks in training/testing
- Still subject to GDPR (legal basis, data subject rights, etc.)
- Balance between utility and privacy
- Appropriate for situations where re-identification may be needed (e.g., research, audits)

Recommendation: Organizations should carefully assess whether data is truly anonymized or only pseudonymized; seek legal/technical advice

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3.3.6 Data Subject Rights in AI Context

GDPR Rights applicable to AI:

Right to Access (Article 15):

- Obtain confirmation if personal data processed
- Access to the data
- Information about processing (purposes, categories, recipients, retention)

AI Application: Right to know if data used by AI, access that data

Right to Rectification (Article 16):

- Correct inaccurate data

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- Complete incomplete data

AI Application: Correct training/operational data

Right to Erasure ("Right to be Forgotten") (Article 17):

- Delete personal data under certain conditions (e.g., no longer necessary, consent withdrawn, unlawful processing)

AI Challenge: Deleting data from trained models (model retraining?)

Right to Restriction of Processing (Article 18):

- Limit processing under certain conditions

Right to Data Portability (Article 20):

- Receive personal data in structured, commonly used, machine-readable format
- Transmit to another controller

Right to Object (Article 21):

- Object to processing based on legitimate interests or direct marketing
- Object to profiling

AI Application: Object to being profiled by AI

Right to Intervention in Automated Decision-Making (Article 22(3)):

- Right to obtain human intervention
- Express point of view
- Contest decision
- Obtain explanation

AI Implementation:

- Build mechanisms for data subject rights into AI systems
- Enable data access, correction, deletion
- Provide explanations for AI decisions
- Human review processes

3.4 Practical Privacy Recommendations for AI

1. Cybersecurity Controls:

- Protect AI systems and data from breaches
- Encryption, access controls, monitoring
- Adversarial attack protections

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2. Privacy by Design and Default (Article 25):

- Embed privacy protections from start
- Default settings maximize privacy
- Data minimization by default

3. Data Governance:

- Clear policies for data collection, use, retention
- Data quality management
- Data lineage and provenance tracking

4. Technical Safeguards:

- Privacy-enhancing technologies (PETs)
- Differential privacy, federated learning, homomorphic encryption, de-identification

5. Data Destruction/Deletion Plans:

- Retain data only as long as necessary
- Secure deletion when no longer needed
- Procedures for data subject deletion requests

6. Privacy Impact Assessments (PIAs/DPIAs):

- Conduct before high-risk processing
- Assess and mitigate privacy risks
- Document and review regularly

7. Technical Documentation and Compliance:

- Document data flows, processing activities
- Records of processing activities (Article 30)
- Audit trails

8. Purpose Specification:

- Define and document purposes for AI data use
- Limit use to specified purposes
- New purposes require new legal basis/consent

9. Human Oversight and Review:

- Human-in-the-loop for high-stakes decisions
- Enable human review of automated decisions
- Appeal mechanisms

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10. Transparency:

- Clear privacy notices
- Explain AI use and data processing
- Accessible information for data subjects

11. Communication with Privacy Authorities:

- Engage with Data Protection Officers
- Report breaches (72 hours under GDPR)
- Consult supervisory authorities when required (DPIA with high residual risk)

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4. OTHER EXISTING LAWS APPLICABLE TO AI

4.1 Intellectual Property Law

AI raises novel IP questions that existing law is still addressing:

4.1.1 Copyright

Key Questions:

1. Can AI outputs be copyrighted?

Issue: Copyright law typically requires human authorship

Current U.S. Position:

- **U.S. Copyright Office**: Works created by AI without human authorship cannot be copyrighted
- Requires "creative expression of a human mind"
- AI-assisted works with significant human creative contribution may be copyrightable

Global Variation:

- Some jurisdictions considering copyright for AI outputs
- Debate ongoing

Practical Implications:

- Pure AI-generated content may be public domain
- Uncertainty affects value of AI-created works (art, music, writing)
- Contracts should address ownership of AI outputs

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2. Is training AI on copyrighted material legal?

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Issue: AI training often uses copyrighted text, images, code scraped from internet

Arguments For (Fair Use):

- Transformative use (learning patterns, not reproducing)
- Public benefit (advancing AI technology)
- Minimal market harm (not substitute for originals)

Arguments Against:

- Copying without permission or payment
- Potential to generate outputs similar to training data
- Harm to creators' markets

Current Status:

- **No settled legal answer globally**
- Active litigation (e.g., lawsuits against OpenAI, Stability AI, GitHub Copilot)
- Varies by jurisdiction (fair use in U.S., different doctrines in EU)

EU Approach:

- Text and Data Mining (TDM) exceptions in Copyright Directive
- Rights holders can opt out (Article 4(3))
- EU AI Act requires GPAI providers to have copyright compliance policy (Article 53(1)©)

Practical Recommendations:

- **License training data** when possible (negotiate with rights holders)
- **Respect opt-outs** (robots.txt, TDM reservations)
- **Use licensed datasets** (commercially available training data)
- **Document data sources** and legal basis
- **Contractual clarity** on data rights, warranties, indemnification in AI agreements

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4.1.2 Patents

Key Question: Can AI be an inventor on a patent?

Thaler v. Vidal (U.S. Courts):

Case: Dr. Stephen Thaler sought to list his AI system ("DABUS") as inventor on patent applications

Holdings:

- **U.S. District Court**, **Federal Circuit**, **Supreme Court** (declined to hear): **AI cannot be inventor**

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- Patent law requires inventor to be "individual" / natural person
- Only humans can be inventors

Rationale:

- Patent statute uses terms implying natural persons ("whoever," "individual")
- Inventorship rights tied to humans (assignment, oaths)
- Public policy (incentivizing human innovation)

Similar Decisions:

- **UK**, **EU**, **Australia** courts: AI cannot be inventor

Practical Implications:

- Human must be listed as inventor (even if AI assisted)
- Organizations should document human involvement in AI-assisted inventions
- Ownership of AI-generated inventions flows from human inventors/employment agreements

Open Question: Patentability of inventions created with AI assistance where human involvement is minimal

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4.1.3 Trade Secrets

Application to AI:

- AI algorithms, training data, model architectures can be trade secrets
- Protection requires secrecy and reasonable efforts to maintain secrecy

Risks:

- Model extraction attacks (stealing AI via queries)
- Reverse engineering
- Insider threats

Protection Measures:

- Access controls and security
- Confidentiality agreements (NDAs)
- Query rate limiting (prevent extraction)
- Watermarking
- Legal agreements with vendors, employees

4.2 Non-Discrimination Laws

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AI systems can violate anti-discrimination laws if they produce discriminatory outcomes:

4.2.1 Employment Discrimination

Applicable Laws (U.S.):

Title VII of Civil Rights Act of 1964:

- Prohibits employment discrimination based on race, color, religion, sex, national origin
- Applies to hiring, firing, promotion, compensation, etc.

Age Discrimination in Employment Act (ADEA): Protects individuals 40+ from age discrimination

Americans with Disabilities Act (ADA): Prohibits discrimination based on disability

Equal Employment Opportunity Commission (EEOC):

- Enforces federal employment discrimination laws
- Guidance on AI in employment
- Employers liable for discriminatory AI (even if unintentional)

AI Applications in Employment:

- Resume screening
- Candidate ranking
- Hiring recommendations
- Performance evaluations
- Promotion decisions
- Termination decisions

Legal Standards:

Disparate Treatment: Intentional discrimination (treating people differently based on protected characteristic)

Disparate Impact: Facially neutral practice that disproportionately affects protected group without business justification

- AI can cause disparate impact through biased training data or flawed algorithms
- Employer must show practice is job-related and consistent with business necessity
- Plaintiff can show less discriminatory alternative exists

NYC Local Law 144:

- Bias audits required for automated employment decision tools (covered in Module 6)
- Employers must publish audit results

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- Candidates must be notified
- Model for other jurisdictions

Challenges with AI Discrimination:

- **Identifying bias**: Requires testing, demographic data (privacy tensions)
- **Data access**: Employers may lack access to vendor AI internals
- **Subjectivity**: What counts as "fair" can be debated
- **Proving causation**: Linking disparate outcomes to AI decisions

Practical Recommendations:

- **Bias testing** before deployment and ongoing
- **Diverse training data**
- **Fairness metrics** and monitoring
- **Human oversight** of AI employment decisions
- **Documentation** of AI use, testing, decision-making
- **Vendor due diligence** (if using third-party AI)
- **Transparency** to applicants about AI use

4.2.2 Credit and Lending Discrimination

Applicable Laws (U.S.):

Equal Credit Opportunity Act (ECOA):

- Prohibits credit discrimination based on race, color, religion, national origin, sex, marital status, age, receipt of public assistance
- Applies to lenders, creditors

Fair Credit Reporting Act (FCRA):

- Regulates consumer reporting agencies
- Accuracy, fairness, privacy of consumer information
- Adverse action notices (must explain credit denial reasons)

Fair Housing Act (FHA):

- Prohibits discrimination in housing (including mortgage lending)
- Protected classes: race, color, religion, national origin, sex, disability, familial status

Regulators:

- **CFPB** (Consumer Financial Protection Bureau)

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- **Federal Reserve**
- **FDIC**, **OCC** (banking regulators)

AI in Credit Decisions:

- Credit scoring models
- Loan approval algorithms
- Risk assessment
- Fraud detection

Adverse Action Notice Requirement (ECOA):

- If credit denied or terms less favorable, creditor must provide reasons
- **Challenge with AI**: "Black box" models difficult to explain
- Regulators expect meaningful, specific reasons (not just "AI score")

Explainability Expectations:

- CFPB and Federal Reserve have indicated AI credit models must be explainable
- Balance between model complexity and transparency
- May limit use of opaque models or require post-hoc explanation techniques

Practical Recommendations:

- Fairness testing across protected groups
- Explainable models or explanation techniques (LIME, SHAP)
- Document decision factors
- Adverse action notices with specific, understandable reasons
- Human review for borderline or contested cases
- Monitor for disparate impact

4.2.3 Other Discrimination Contexts

Insurance:

- State laws prohibit unfair discrimination in insurance underwriting, pricing
- AI risk assessment must not discriminate based on protected characteristics
- Actuarial justification required for rate differences

Housing:

- Fair Housing Act (as above)
- AI in tenant screening, rent pricing

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Healthcare:

- Section 1557 of Affordable Care Act: Prohibits discrimination in healthcare based on race, color, national origin, sex, age, disability
- AI diagnostics, treatment recommendations must not discriminate

Education:

- Title VI (race), Title IX (sex), Section 504 (disability)
- AI in admissions, grading, student assessment

Public Services:

- Constitutional equal protection (U.S.)
- AI used by government must not discriminate

4.3 Consumer Protection Law

General Principle: Existing consumer protection laws apply to AI; no AI exemption

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4.3.1 U.S. Federal Trade Commission (FTC)

Authority: FTC Act Section 5—Prohibits "unfair or deceptive acts or practices"

Application to AI:

Deceptive Practices:

- **False claims** about AI capabilities (overstating accuracy, performance)
- **Misleading statements** about AI use (claiming human when AI, vice versa)
- **Fake reviews** or content (AI-generated deceptive content)
- **Deepfakes** used deceptively in advertising

Examples of FTC Enforcement:

- Claims that AI is more accurate than it is
- Failing to disclose AI use when material to consumers
- AI-generated fake testimonials

Unfair Practices:

- Practices causing substantial harm to consumers
- Harm not outweighed by benefits
- Harm not reasonably avoidable by consumers

AI Examples:

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- **Algorithmic bias** causing discriminatory outcomes
- **Privacy violations** through excessive AI data collection
- **Security failures** in AI systems leading to breaches
- **Manipulative design** using AI to exploit consumers

FTC Guidance and Statements:

- FTC has issued guidance stating existing rules apply to AI
- Focus on truth in advertising, privacy, security, fairness
- Warnings about "AI washing" (exaggerating AI use or capabilities)

Practical Recommendations:

- **Truthful claims**: Don't overstate AI capabilities
- **Disclose AI use**: Be transparent when material to consumers
- **Substantiation**: Have evidence to back up AI performance claims
- **Fairness**: Test for and mitigate bias
- **Privacy and security**: Protect consumer data
- **Avoid deception**: Don't use AI to deceive (fake content, reviews)

4.3.2 Other Jurisdictions

EU:

- Unfair Commercial Practices Directive
- Consumer Rights Directive
- AI-specific consumer protections in EU AI Act (transparency, disclosure)

Other Countries: Similar consumer protection frameworks apply to AI

4.4 Product Liability Law

Issue: Who is liable when AI causes harm?

Relevance: As AI deployed in products (autonomous vehicles, medical devices, consumer products), liability questions arise

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4.4.1 Liability Regimes

Two Main Regimes:

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Fault-Based Liability (Negligence):

Principle: Liable if failed to exercise reasonable care

Elements:

- Duty of care
- Breach of duty (failure to meet standard)
- Causation (breach caused harm)
- Damages

AI Context:

- Did developer/deployer act reasonably in designing, testing, monitoring AI?
- Standard of care evolving with AI best practices

Challenges:

- Proving negligence (what is "reasonable" for AI?)
- Establishing causation (AI complexity, opacity)

Strict Liability:

Principle: Liable for harm caused by defective product, regardless of fault

Elements (U.S. common law):

- Product was defective (design, manufacturing, warning defect)
- Defect existed when left manufacturer's control
- Defect caused harm

AI Context:

- Is AI software a "product" for strict liability? (Unsettled—historically software treated differently)
- What constitutes a "defect" in AI? (Fails to meet expectations? Bias? Error?)

EU Approach:

- Traditional product liability focuses on physical products
- EU revising directives to address AI and digital products

4.4.2 AI-Specific Liability Challenges

Attributing Harm:

- **Complexity**: Hard to trace decisions in complex AI

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- **Opacity**: Black-box models obscure causation
- **Autonomy**: AI acts independently; who responsible?
- **Multiple parties**: Developer, deployer, data provider—who liable?

Probabilistic Nature:

- AI errors inevitable (no 100% accuracy)
- When is error a "defect" vs. acceptable imperfection?

Adaptation and Learning:

- AI behavior changes after deployment
- Developer may not foresee post-deployment behavior

4.4.3 Emerging Legal Developments

EU AI Liability Directive (Proposed):

New Proposals (under development):

- **AI Liability Directive**: New fault-based liability rules for AI harm
- **Updated Product Liability Directive**: Extend strict liability to software and AI

Key Features:

- **Presumption of causal link**: If provider/deployer non-compliant and harm plausibly linked, causation presumed (shifts burden)
- **Right to disclosure**: Victims can request disclosure of AI technical documentation
- **Easier compensation**: Reduce burdens on victims

Purpose: Address challenges in proving AI-related harm; facilitate victim compensation

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U.S. Developments:

Current State:

- Product liability law varies by state
- Application of strict liability, negligence, breach of warranty depends on jurisdiction and facts
- **Unsettled**: Whether AI software is "product" for strict liability

Debate:

- Some argue AI should be treated as product → strict liability
- Others: software traditionally not subject to strict product liability
- Ongoing litigation and legislative activity

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4.4.4 Practical Liability Risk Management

For Providers/Deployers:

1. Robust Development and Testing:

- Follow best practices (Module 4)
- Comprehensive testing (accuracy, safety, robustness, bias)
- Documentation of development process

2. Clear Warnings and Instructions:

- Communicate AI limitations
- Provide clear instructions for safe use
- Warn about inappropriate uses

3. Post-Market Monitoring:

- Continuous monitoring for issues
- Incident response and remediation
- Updates and patches

4. Insurance:

- Liability insurance for AI risks
- Cyber insurance
- Professional liability insurance

5. Contractual Protections:

- Liability allocation in contracts (vendor agreements, terms of service)
- Indemnification clauses
- Disclaimers (where legally enforceable)
- Limitation of liability

6. Compliance:

- Meet regulatory requirements (EU AI Act, sector-specific regulations)
- Demonstrates due diligence in liability defense

7. Documentation:

- Maintain records of design choices, testing, risk assessments
- Demonstrate reasonable care

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5. COMPLIANCE STRATEGIES FOR MULTI-JURISDICTIONAL ORGANIZATIONS

5.1 Challenges

No One-Size-Fits-All:

- Different legal regimes across jurisdictions
- Conflicting requirements possible
- Complexity in managing multiple compliance programs

Resource Constraints:

- Separate compliance for each jurisdiction expensive and inefficient
- Need for scalable approach

5.2 Recommended Strategy: Harmonize to Strictest Requirements

Approach:

1. **Identify all applicable jurisdictions** (where you operate, where customers are, where data subjects reside)
2. **Understand requirements in each** (regulations, standards, best practices)
3. **Identify strictest requirements** across jurisdictions
4. **Develop unified compliance framework** based on highest standards
5. **Implement globally** (or across all relevant operations)

Rationale:

- **Efficiency**: One framework, not multiple
- **Risk mitigation**: Meet highest bar → likely compliant everywhere
- **Simplicity**: Easier to manage and audit
- **Reputation**: Demonstrates commitment to highest standards

Example:

- If operating in EU and U.S., EU AI Act may be strictest
- Develop governance meeting EU AI Act → likely complies with U.S. requirements and best practices

Adjustments:

- Fine-tune for jurisdiction-specific requirements (e.g., local language, specific reporting)
- But core framework unified

5.3 Leverage Existing Privacy and Data Protection Processes

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Integration:

- AI governance shouldn't be siloed
- Leverage existing processes (DPIAs, privacy by design, data governance)
- Extend and adapt for AI-specific considerations

Benefits:

- Efficiency (build on what exists)
- Consistency (aligned approach)
- Expertise (leverage existing privacy/data protection knowledge)

Example:

- DPIA process for privacy → expand to include AI-specific risks (bias, safety, etc.) → becomes AIA
- Existing risk management → integrate AI risks

5.4 Stay Alert and Prepared to Adjust

Dynamic Landscape:

- AI regulation evolving rapidly
- New laws, guidance, enforcement actions
- Technology advancing

Ongoing Activities:

- **Monitor regulatory developments** (subscribe to updates, engage with industry groups)
- **Review and update** governance frameworks regularly
- **Training and awareness** for staff on new requirements
- **Flexibility** in governance (adaptable to changes)
- **Engage with regulators** (seek guidance, participate in consultations)

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6. KEY DEFINITIONS FOR MODULE 7

ISO 22989: International standard establishing terminology and concepts for AI.

ISO 42001: International standard for AI management systems; guidance for implementing AI governance.

HUDERIA: Council of Europe framework for Human Rights, Democracy, and Rule of Law Impact Assessment for AI.

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Fair Information Practices (FIPs): Foundational privacy principles (notice, choice, access, integrity, enforcement).

GDPR (General Data Protection Regulation): EU comprehensive data protection law.

Article 22 (GDPR): Right not to be subject to solely automated decision-making with significant effects.

Special Category Data: Sensitive personal data under GDPR Article 9 (race, religion, health, biometric, etc.).

Anonymization: Irreversible removal of identifiers; data no longer personal under GDPR.

Pseudonymization: Reversible replacement of identifiers with pseudonyms; still personal data under GDPR.

DPIA (Data Protection Impact Assessment): Privacy risk assessment required under GDPR for high-risk processing.

Data Minimization: Privacy principle to collect only necessary data.

Purpose Limitation: Privacy principle to use data only for stated purposes.

Thaler v. Vidal: U.S. case ruling AI cannot be inventor for patents; only humans can be inventors.

Copyright: Legal protection for original works of authorship; AI-generated content not copyrightable without human authorship (U.S.).

Trade Secret: Confidential business information protected by secrecy; applicable to AI algorithms, models.

Title VII: U.S. law prohibiting employment discrimination based on race, color, religion, sex, national origin.

EEOC (Equal Employment Opportunity Commission): U.S. agency enforcing employment discrimination laws.

NYC Local Law 144: Requires bias audits for automated employment decision tools in NYC.

ECOA (Equal Credit Opportunity Act): U.S. law prohibiting credit discrimination.

FCRA (Fair Credit Reporting Act): U.S. law regulating consumer reporting, accuracy, adverse action notices.

FTC (Federal Trade Commission): U.S. agency enforcing consumer protection laws; applies to AI (unfair/deceptive practices).

Fault-Based Liability (Negligence): Liability for failing to exercise reasonable care.

Strict Liability: Liability for harm from defective product regardless of fault.

Product Liability: Legal responsibility for harm caused by products; application to AI unsettled.

AI Liability Directive (EU Proposed): Proposed EU rules for easier victim compensation in AI harm cases (presumption of causation, disclosure rights).

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Harmonize to Strictest Requirements: Compliance strategy of developing unified framework based on most stringent applicable law.

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7. STUDY TIPS FOR MODULE 7

Key Concepts to Master:

1. ✅ Understand **global AI regulation landscape** (EU, U.S., China, Canada, Singapore, Brazil, India, Japan)
2. ✅ Know **key U.S. state laws** (California AB 2013/SB 942, Colorado SB 24-205, NYC Local Law 144)
3. ✅ Master **ISO standards** (ISO 22989 terminology, ISO 42001 management system)
4. ✅ Understand **GDPR Article 22** (automated decision-making) and requirements
5. ✅ Know **special category data** (Article 9) and exceptions for processing
6. ✅ Distinguish **anonymization vs. pseudonymization** and GDPR treatment
7. ✅ Understand **Thaler v. Vidal** (AI cannot be patent inventor)
8. ✅ Know **non-discrimination laws** (Title VII, EEOC, ECOA, FCRA) and AI application
9. ✅ Master **FTC authority** over AI (unfair/deceptive practices)
10. ✅ Understand **product liability** challenges with AI and emerging EU/U.S. approaches

Common Pitfalls to Avoid:

- ❌ Confusing AI-specific laws (EU AI Act) with privacy laws (GDPR)—both apply to AI
- ❌ Not understanding that pseudonymized data is still personal data under GDPR
- ❌ Forgetting Article 22 exceptions (contract, law, explicit consent)
- ❌ Not knowing copyright status of AI-generated content (not copyrightable in U.S. without human authorship)
- ❌ Confusing fault-based liability with strict liability
- ❌ Not understanding FTC applies existing consumer protection laws to AI

Practical Application:

- Map applicable laws to a hypothetical AI system in multiple jurisdictions
- Design DPIA process for AI system processing personal data
- Develop copyright compliance policy for GPAI (as required by EU AI Act Article 53)
- Create fairness testing process addressing non-discrimination laws
- Draft adverse action notice for credit AI decision (ECOA/FCRA)

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- Design multi-jurisdictional compliance strategy using harmonization approach

Memorization Aids:

ISO Standards (22-42):

- **ISO 22989**: AI **terminology** (22 → concepts)
- **ISO 42001**: AI **management system** (42 → governance)

GDPR Key Articles for AI:

- **Article 9**: Special category data
- **Article 22**: Automated decision-making
- **Article 25**: Privacy by design/default
- **Article 35**: DPIA

Article 22 Exceptions (C-L-C):

- **C**ontract (necessary for)
- **L**aw (authorized by)
- **C**onsent (explicit)

Special Category Data (Article 9 - R-P-R-T-G-B-H-S):

- **R**acial/ethnic origin
- **P**olitical opinions
- **R**eligious/philosophical beliefs
- **T**rade union membership
- **G**enetic data
- **B**iometric data (for identification)
- **H**ealth data
- **S**ex life/sexual orientation

U.S. Employment Discrimination Laws (T-A-A):

- **T**itle VII (race, color, religion, sex, national origin)
- **A**DEA (age 40+)
- **A**DA (disability)

U.S. Credit/Lending Laws (E-F-F):

- **E**COA (Equal Credit Opportunity Act)
- **F**CRA (Fair Credit Reporting Act)
- **F**HA (Fair Housing Act—for mortgage lending)

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Liability Regimes (F-S):

- **F**ault-based (negligence—reasonable care)
- **S**trict liability (product defect—no fault required)

FIPs (Five Principles - N-C-A-I-E):

- **N**otice/Awareness
- **C**hoice/Consent
- **A**ccess/Participation
- **I**ntegrity/Security
- **E**nforcement/Redress

Compliance Strategy (I-U-I-D-I):

- **I**dentify jurisdictions
- **U**nderstand requirements
- **I**dentify strictest
- **D**evelop unified framework
- **I**mplement globally

End of Module 7 Notes

🎉 **CONGRATULATIONS!** 🎉

You now have **comprehensive study notes for all 7 AIGP modules**!

Complete AIGP Study Guide Summary:

✅ **Module 1**: Foundations of Artificial Intelligence
✅ **Module 2**: AI Impacts on People and Responsible AI Principles
✅ **Module 3**: Responsible AI Governance and Risk Management
✅ **Module 4**: Governing AI Development
✅ **Module 5**: Governing AI Deployment
✅ **Module 6**: The EU AI Act
✅ **Module 7**: Other Laws and Standards Related to AI

Final Exam Preparation Tips:

Study Approach:

1. **Review all 7 module notes systematically**

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2. **Practice with the 30-question tests** for each module (210 questions total)
3. **Focus on connections** between modules (e.g., Module 3 frameworks applied in Modules 4-5)
4. **Memorize key definitions, lists, frameworks** using the mnemonics provided
5. **Understand real-world application** through case studies and examples
6. **Review practical scenarios** and how to apply concepts

High-Priority Topics (Frequently Tested):

- NIST AI RMF (Four Functions, Seven Characteristics)
- EU AI Act (Risk categories, Obligations, Timeline, Penalties)
- Types of bias and mitigation strategies
- Privacy-enhancing technologies (PETs)
- Data governance (Five V's, quality, lineage)
- Testing types (accuracy, robustness, bias, adversarial, etc.)
- Three Lines of Defense
- Risk Mitigation Hierarchy
- GDPR and AI (Article 22, DPIAs, special category data)
- Deployment environments and monitoring

Cross-Module Integration:

- **Modules 1-2**: Foundation → understand AI and its impacts
- **Module 3**: Governance structure → how to organize
- **Modules 4-5**: Lifecycle governance → development and deployment
- **Modules 6-7**: Legal compliance → regulations and laws

Time Management:

- Allocate study time proportionally to module length and complexity
- Modules 4, 5, 6 are longer and more detailed—spend more time
- Practice time-bound test-taking with the 30-question tests

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